First Demonstration of Airborne MIMO SAR System for Multimodal Operation

Jie Wang, Xing-Dong Liang, Long-Yong Chen, and Yan-Lei Li

***Abstract*— Synthetic aperture radar (SAR) systems have extensively contributed to diverse scientific applications in the recent past decades. For the next generation of top-level SAR systems, the multimodal operation of wide swath imaging, sliding spotlight, and highly sensitive ground moving target indication (GMTI) is becoming increasingly crucial. Following this concept, in this article, a novel airborne multiple-input–multiple-output (MIMO) SAR system, along with the space–time coding (STC) waveform scheme and the phased array, is developed to realize the multimodal operation. In particular, the airborne MIMO SAR system consists of two transmitters and four receivers. The signals of the multiple modes, which overlap in the frequency domain, are interpulse modulated and transmitted by different subarrays concomitantly. The superposed echoes are separated by range-Doppler filtering. Based on the separated echoes, we can obtain eight equivalent transceiver channels and, as a result, enhance the signal-to-noise ratio (SNR) for airborne multimodal operation. In the future, the STC waveform scheme and flight experiments of this novel airborne MIMO SAR system will be further investigated to reduce the pulse repeat frequency (PRF) for spaceborne multimodal operation.**

***Index Terms*— Airborne multiple-input–multiple-output (MIMO) synthetic aperture radar (SAR), multimodal operation, space–time coding (STC) waveform scheme.**

I. Introduction

SYNTHETIC aperture radar (SAR) is a well-proven imaging technique for the remote sensing of Earth [1]–[3]. Expanding the performance of SAR systems, capabilities, such as wide coverage, fine geometric resolution, and high observation efficiency, have become increasingly crucial for future SAR missions [4]–[8]. Ocean traffic monitoring applications, for example, require SAR systems to cover the maximum possible area over sea to grasp the overall distribution of vessels. Nevertheless, a higher geometric resolution is required for the precise observation of suspicious ships. The high-resolution wide swath (HRWS) SAR imaging mode, which can be realized by performing digital beamforming (DBF) on multiple receive antennas [9]–[14], has emerged as a promising method for forthcoming applications. However, in certain cases, there is no need to image all the targets with high resolution. For uninteresting targets, high-resolution driven resources are unnecessary. Consequently, apart from the HRWS SAR modes, multimodal operation, which involves multiple SAR modes such as wide swath, spotlight, and ground moving target indication (GMTI) concomitantly performed on a single system, is becoming increasingly crucial. It can allocate system resources such as bandwidth, power, and apertures to different targets in accordance with the actual situation of the observed area. For example, whereas the wide swath low-resolution mode, which consumes fewer system resources, is used to master the overall situation of the sensing area, the system can simultaneously gather a major share of the resources to precisely observe targets of interest through spotlight modes.

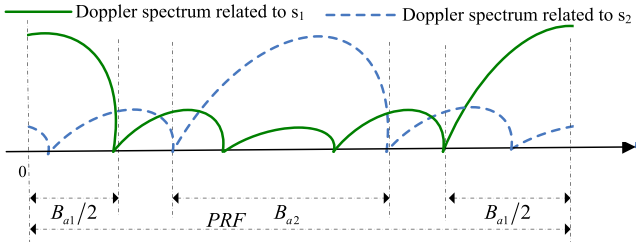


Fig. 1. STC configuration for the multimodal operation.

However, for state-of-the-art SAR systems, the realization of this concept is a great challenge. Conventional single-antenna SAR systems are not suitable due to the restricted degrees of freedom [15]–[17]. Multichannel SAR systems do not support waveform diversities. Therefore, in this article, a novel airborne multiple-input–multiple-output (MIMO) SAR system is developed to realize the concept. The novel MIMO SAR system, which is implemented by adding transmit antennas to multiple receive antennas, can acquire sufficient degrees of freedom (waveform diversities; equivalent phase centers in the range, azimuth, and elevation directions) by transmitting mutually orthogonal waveforms [18]–[26]. While the additional equivalent phase centers can be used to improve the imaging qualities of the classical SAR modes, the waveform diversities can be exploited to realize multimodal operation.

With proper modification, this MIMO SAR system can be further improved as an experimental platform for various investigations on the future remote sensing technologies and missions. For example, in [27], a novel idea of exploiting MIMO diversities to realize the concept of joint wireless

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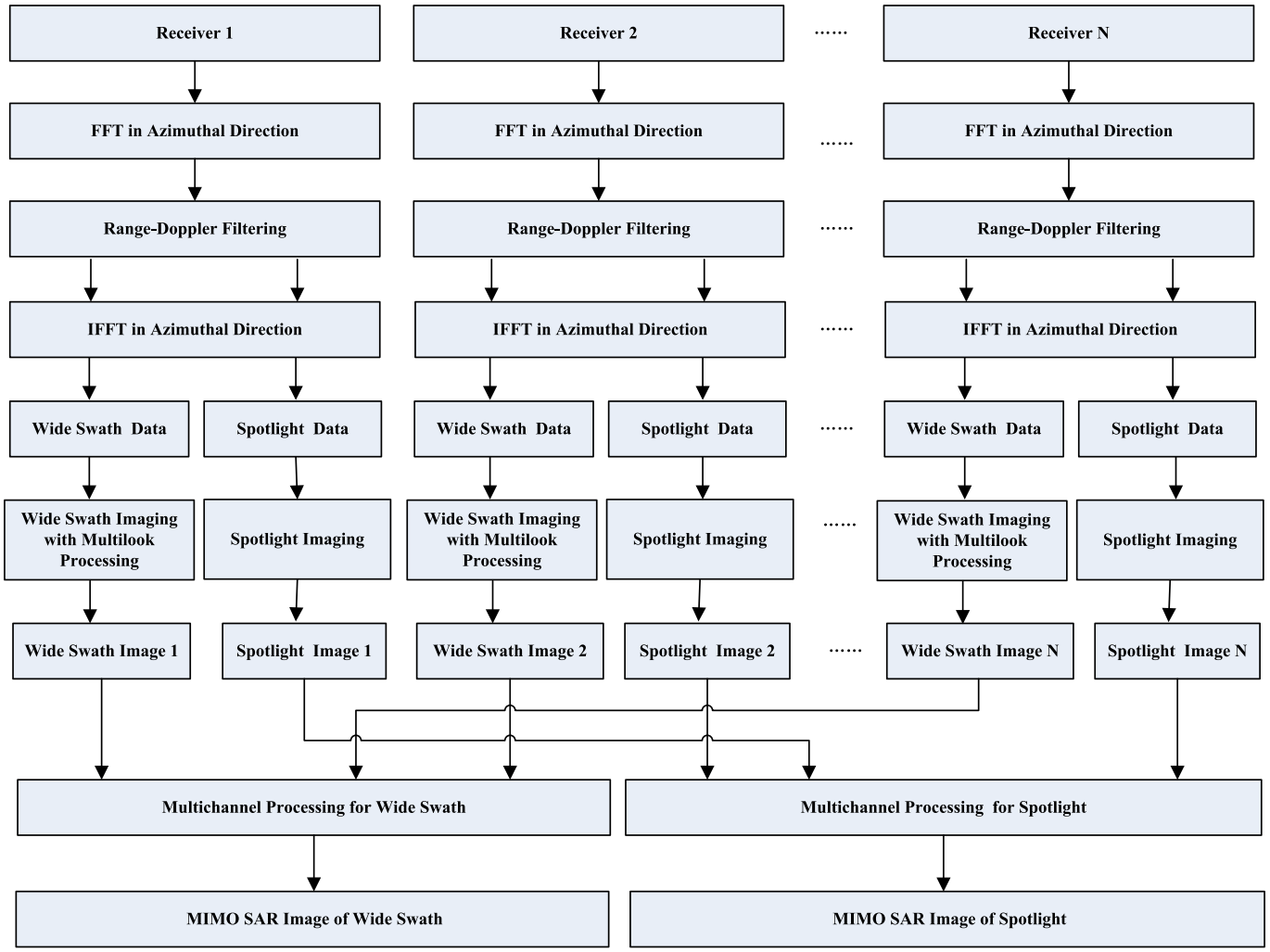


Fig. 2. Multimodal imaging procedure of the airborne MIMO SAR system.

communication and high-resolution SAR imaging is proposed. The MIMO SAR system has been successfully modified as an experimental platform to validate this novel idea. It should be noted that this article is quite distinct from [27]. This article aims to present the intrinsic purpose, significance, technique details, and multimodal SAR results of the MIMO system.

The MIMO concept was proposed by Paulraj and Kailath [28] for the purpose of increasing the channel capacity in wireless communications. It has shown tremendous potential for various applications, including wireless communication and radar sensing. In particular, this multiple antenna concept has triggered intense research on massive MIMO technologies for the next generation of wireless communication. In the SAR community, MIMO technologies are also gaining popularity [29]–[35]. The MIMO SAR, formally proposed by Ender [36], is a promising concept. For MIMO antennas arranged along a track, more spatial samples in the azimuthal direction can be produced to realize HRWS imaging or GMTI. If the antennas are arranged in a cross-track layout, 3-D imaging and multibaseline interferometry can be performed.

Accordingly, several works on MIMO SAR systems have been published in recent decades. However, the performances of these systems are seriously limited by the available sets of orthogonal waveforms [18]. According to the ambiguity function defined by Woodward [37], the energy of the imperfect orthogonal waveforms is spread in the time–frequency plane. The impact of ambiguous energy is negligible for traditional radar systems. However, in regard to SAR applications, ambiguous energy accumulates. The accumulated ambiguous energy, which may exceed the energy of the focused low backscatters, degrades the SAR images significantly.

Certain works on multidimensional waveform schemes, such as short-term shift-orthogonal (STSO) waveforms, space–time coding (STC) waveforms, and orthogonal frequency-division multiplexing (OFDM) chirp waveforms, have emerged as potential solutions to the MIMO SAR ambiguous energy issue [38]–[46]. The essential difference between multidimensional and conventional waveforms is that multidimensional waveforms are modulated in multiple dimensions, including time, space, frequency, Doppler, and polarization. Consequently, although the waveforms of multiple SAR modes share the same time–frequency coverage, their echoes can still be perfectly separated by filtering in other domains. Therefore, the ambiguous energy can be avoided. In this work, the STC waveforms are considered. The remainder of this article is organized as follows. The STC waveform scheme, overviewed

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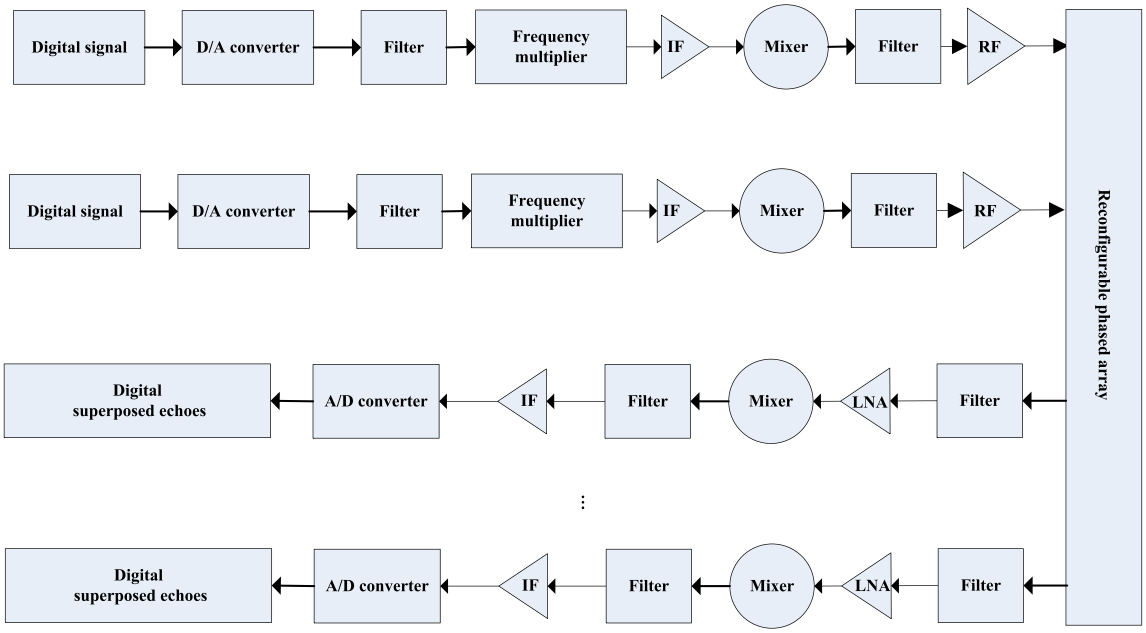


Fig. 3. Signal flow diagram of the airborne MIMO SAR system.

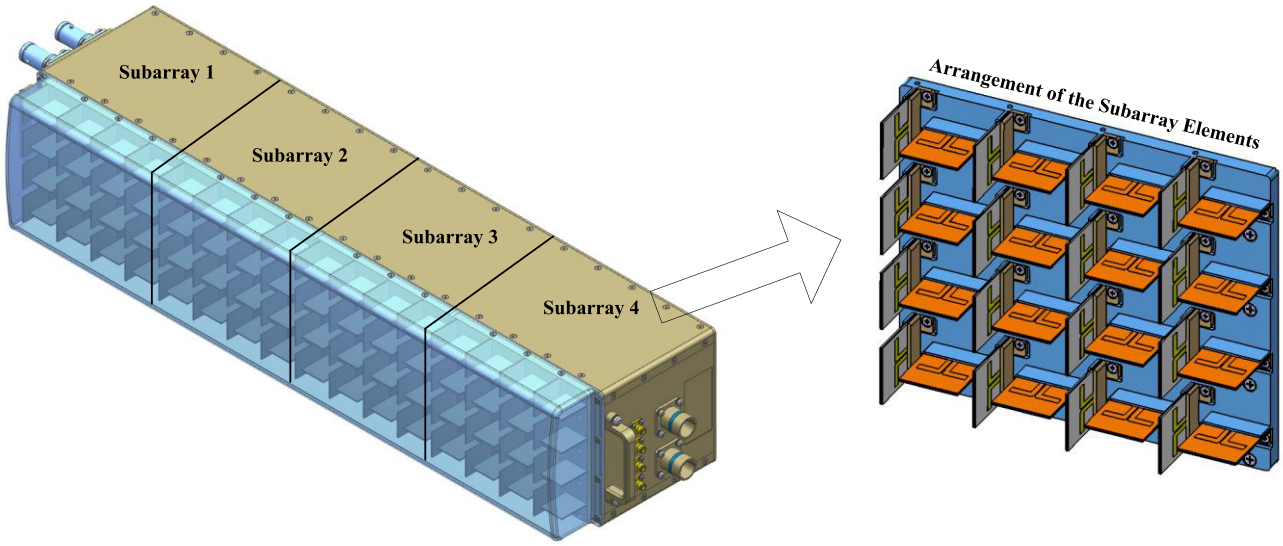


Fig. 4. Structure of the phased array antenna.

in Section II, is followed by the multimodal configuration of the airborne MIMO SAR in Section III. The experimental flight results are detailed in Section IV, and the conclusions are given in Section V.

II. Overview of the STC Waveform Scheme

The STC schemes are essential for multimodal operation. Assuming the knowledge in [45], we give a short overview of the modulation schemes and point out the important properties.

Assuming that there are two transmitters, the signal model of the STC scheme is based on the following matrix:

$$\mathbf{S} = \begin{bmatrix} s_1 & s_2^* \\ s_2 & -s_1^* \end{bmatrix} \tag{1}$$

where $(\cdot)^*$ denotes the complex conjugate operator and s_1 and s_2 denote the basic signals.

By specializing $s_1 = s_2^*$, we can obtain

$$\mathbf{S} = \begin{bmatrix} s_1 & s_1 \\ s_2 & -s_2 \end{bmatrix}. \tag{2}$$

The first and second rows in $\mathbf{S}$ denote the signals transmitted by the first and second transmitters, respectively. The first and second columns in $\mathbf{S}$ denote the signals transmitted at the first and second pulse repeat intervals (PRIs). In particular, s_1 and s_2 are transmitted by the first and second transmitters at the first PRI, following which s_1 and $-s_2$ are radiated by the same antennas. With the increase in the number of PRIs, this coding matrix repeats.

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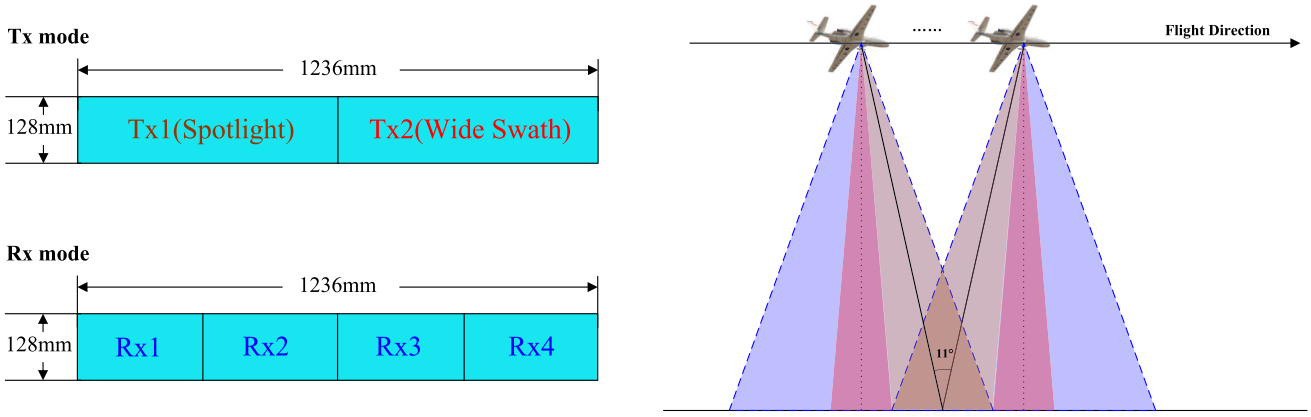


Fig. 5. Antenna configuration and geometry of the multimodal operation concept. For the geometry, the red, orange, and blue sectors denote the wide swath transmitting beams, the spotlight transmitting beams, and the receiving beams, respectively.

For the side-looking geometry, the signals to be transmitted by the first and second transmitters can be rewritten as

$$\begin{aligned} s_1'(t_r, t_a) &= \omega(t_a) s_1(t_r) \\ s_2'(t_r, t_a) &= \omega(t_a) \cdot s_2(t_r) \cdot \exp(jm\pi) \\ &= \omega(t_a) \cdot s_2(t_r) \cdot \exp(j2\pi f_{ac} t_a) \end{aligned} \tag{3}$$

where

$$f_{ac} = \frac{\text{PRF}}{2}, \quad t_a = \frac{m}{\text{PRF}} \tag{4}$$

t_a denotes the slow time, $\omega(t_a)$ is the azimuth envelope, PRF is the pulse repeat frequency, and m is the azimuth sample number.

Due to the linear phase in the azimuthal direction, the superposed echoes are shifted with PRF/2 in the range-Doppler domain (see Fig. 1). It must be noted that the STC model in (2) is based on the assumption that s_1 equals s_2^*. For the HRWS mode, this assumption can be satisfied by employing up and down chirps of the same bandwidths. However, for the spotlight and wide swath modes, the bandwidth requirements of which are generally different, this assumption cannot be satisfied. Thus, to avoid this assumption for the multimodal operation concept, we can start the modulation directly from (3) and (4). To this end, the STC approach becomes an azimuth phase coding (APC) scheme. Its application in the SAR context dates back 2004. Compared with the APC schemes proposed for range ambiguity suppression in spaceborne multichannel SAR and improved for MIMO SAR imaging [47]–[49], the difference is that the coded phases in the previous studies are quadratic relative to m. However, after APC demodulation, the residual phase, which is linear, becomes similar to that in (3). Henceforth, from the viewpoint of azimuth coding, $s_1(t_r)$ and $s_2(t_r)$ in (3) and (4) can be any form of radar signals. They are not required to be conjugate at all. In our experiments, $s_1(t_r)$ and $s_2(t_r)$ are up and down chirps. Their bandwidths can either be the same or different. The echoes related to the two signals can be separated by performing range-Doppler filtering on the superposed echoes received by each antenna individually. However, the PRF must be larger than the sum of the Doppler bandwidths. Moreover, the sidelobes of the Doppler spectra, which primarily result from the azimuth antenna pattern, may introduce residual intramodal interference. Nevertheless, the residual interference can be reduced by exploiting the digital phased arrays. The increased PRF has little impact on the range ambiguities for the airborne SAR geometry.

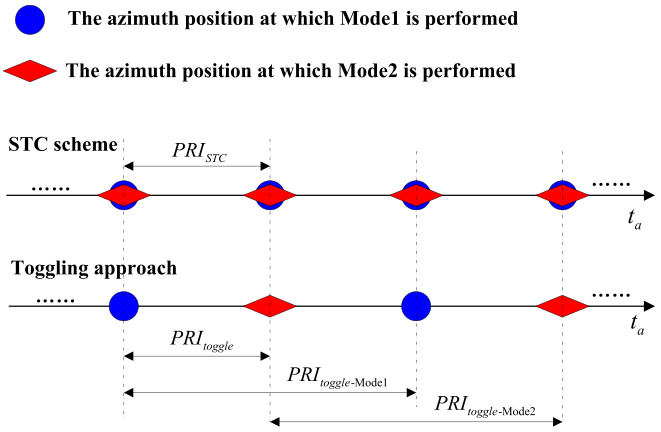


Fig. 6. Comparison between the STC and the toggling schemes in accordance with the distribution of the transceiving positions in the azimuthal direction. PRI_{STC} and $\text{PRI}_{\text{toggle}}$ are the PRIs of the STC and the toggling schemes, respectively. $\text{PRI}_{\text{toggle}}$ equals PRI_{STC}. $\text{PRI}_{\text{toggle-Mode1}}$ and $\text{PRI}_{\text{toggle-Mode2}}$ are the individual PRIs of Mode1 and Mode2 in the toggling scheme, respectively.

With the STC waveform scheme, the processing procedure of the airborne multimodal operation is shown in Fig. 2. The key point of this procedure is demodulation. For each receiver, once the superposed echoes are separated by range-Doppler filtering, the ωK algorithm can be performed on the two separated echoes to obtain one wide swath image and one spotlight image. Then, for each of the modes, we can obtain N images with the employment of N receivers. Based on these multiple images, multichannel processing can be performed by image registration and addition to improve the signal-to-noise ratio (SNR). In addition, since multiple antennas are placed along the track, we can further perform the GMTI mode to detect moving targets.

The multimodal concept of wide swath imaging, spotlight, and GMTI will be validated by the experimental flight results in Section IV. For a comparison, we also performed the HRWS mode in the following experiments. The processing procedure

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(a) (b)

Fig. 7. Phased array loaded on the plane. (a) Front of the phased array. (b) Back of the phase array.

of the HRWS mode is similar to that in Fig. 2. The difference is that the wide swath mode and the spotlight mode in Fig. 2 become the same one, namely, the HRWS mode. Then, for each receiver, we can obtain two images. The SNR of the HRWS can be improved by adding the *2N* images.

Generally, for spaceborne geometry, the realization of the HRWS mode is limited by the fundamental antenna constraint. If one tries to realize spaceborne HRWS imaging, the echoes should be rearranged according to the distribution of the equivalent phase centers in the azimuthal direction to reduce the system PRF without degrading the azimuthal resolution. However, for airborne geometry, the realization of HRWS imaging is mainly limited to the available SNR. Since the synthesis of the *2N* images can improve the SNR, the proposed processing procedure is qualified for airborne HRWS imaging as well. Moreover, the system PRF, which must be larger than twice the Doppler bandwidth, is to ensure a better separation performance for the STC scheme. It has little impact on the range ambiguities of the airborne geometry.

III. Multimodal Configuration

The signal flow diagram of the airborne MIMO SAR system is shown in Fig. 3. The reconfigurable system mainly consists of two transmitting channels and four receiving channels. The STC waveforms are stored in the flash as digital form. In the transmitting channels, the waveforms are D/A converted into analog, upconverted to the C-band, and subsequently radiated by the phased array. The echoes captured by the array are downconverted, amplified, and sampled by the A/D converter. The multimodal configuration is as follows.

The phased array is essential for the multimodal operation of wide swath imaging, spotlight, and GMTI. It consists of four subarrays. There are 16 horizontally and 16 vertically polarized elements in each subarray. The structure of the phased array antenna is shown in Fig. 4. In particular, the horizontally and vertically polarized elements are interleaved in the azimuthal direction. For both polarizations, there are four elements in the azimuth and four elements in the elevation. The distances between the elements in the azimuth and elevation are 38 and 32 mm, respectively. Thus, the subarray is capable of scanning $\pm 20^\circ$ and $\pm 15^\circ$ in the azimuth and elevation, respectively. With the scanning ability, we are able to perform the spotlight mode.

The phased array is divided into two subapertures during the transmission of signals and reconfigured as four subapertures while receiving the multimodal echoes. The 3-dB beamwidths in the azimuthal direction for the transmitting and receiving subapertures are designed as 5.5° and 11°, respectively. Based on the phased array, the sidelobes of the antenna pattern in the azimuthal direction can be suppressed by designing the coefficients of the elements. Thus, if the STC waveform scheme is employed to realize multimodal operation, we can ensure that the residual interference introduced by the sidelobes of the Doppler spectra can be neglected in the airborne MIMO SAR.

Taking the STC scheme and the phased array into consideration, the antenna configuration and the geometry of the multimodal operation are shown in Fig. 5. In particular, Tx1 and Tx2, which transmit $s_1'(t_r, t_a)$ and $s_2'(t_r, t_a)$, are used for the spotlight and wide swath modes, respectively. The scanning angle of Tx1 is $\pm 5.5^\circ$ in the azimuthal direction. There is no steering for Tx2, Rx1, Rx2, Rx3, and Rx4. Thus, theoretically speaking, with the scanning angle in the azimuthal direction, we can obtain an azimuthal resolution of 0.14 m for the spotlight mode. In addition, since the main lobe of the spotlight transmitting beam is covered by the

TABLE I
Parameters and Performances of Multimodal Operation

System parameters	Carrier Frequency	5.4 GHz
	Antenna Size	1236 mm×128 mm
	Phase Centers	8
	Transmitting Power	$\leqslant$1800 W
	Platform Height	8 km
	Platform Velocity	150 m/s
	System PRF	2000 Hz
	Intra-interferences	-35 dB
Wide swath mode	Bandwidth	100 MHz
	Duration	150 μs
	Doppler Bandwidth	648 Hz
	Range Resolution	1.54 m
	Azimuthal Resolution	0.91 m
	Swath Width	50.35 km
Spotlight mode	Bandwidth	560 MHz
	Duration	150 μs
	Doppler Bandwidth	1035 Hz
	Range Resolution	0.29 m
	Azimuthal Resolution	0.15 m
GMTI mode	Minimum Detected Velocity	1.69 m/s

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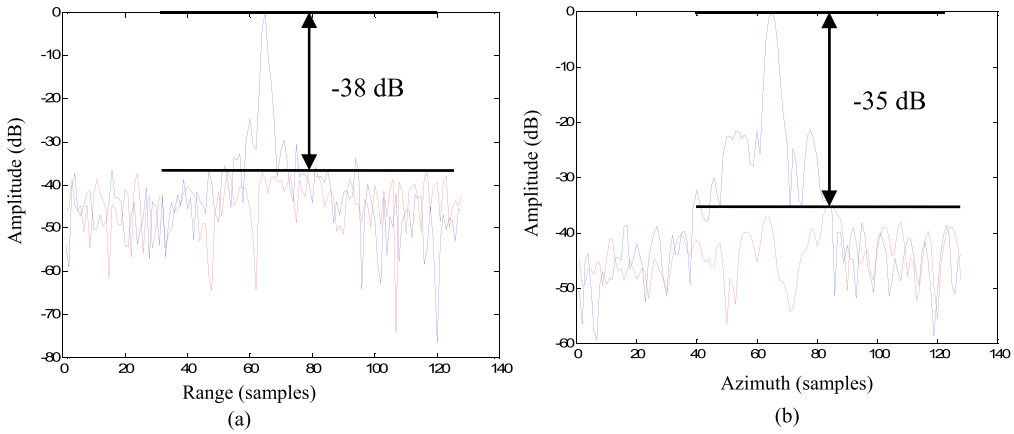


Fig. 8. Intrainterferences introduced by the sidelobes of the Doppler spectra. (a) Range slice of the corner reflector. (b) Azimuth slice of the corner reflector. The blue lines are the slices of the corner reflector in the image that the pulse compression in the range is realized by matched-filtering the Doppler baseband echoes with s1. The red lines are the slices of the same corner reflector in the image that the pulse compression in the range is realized by matched-filtering the Doppler high band echoes with s1. Since there is no peak in the red line and the blue peak to red sidelobe ratio is-35 dB, it can be concluded that the intrainterferences introduced by the sidelobes of the Doppler spectra are negligible.

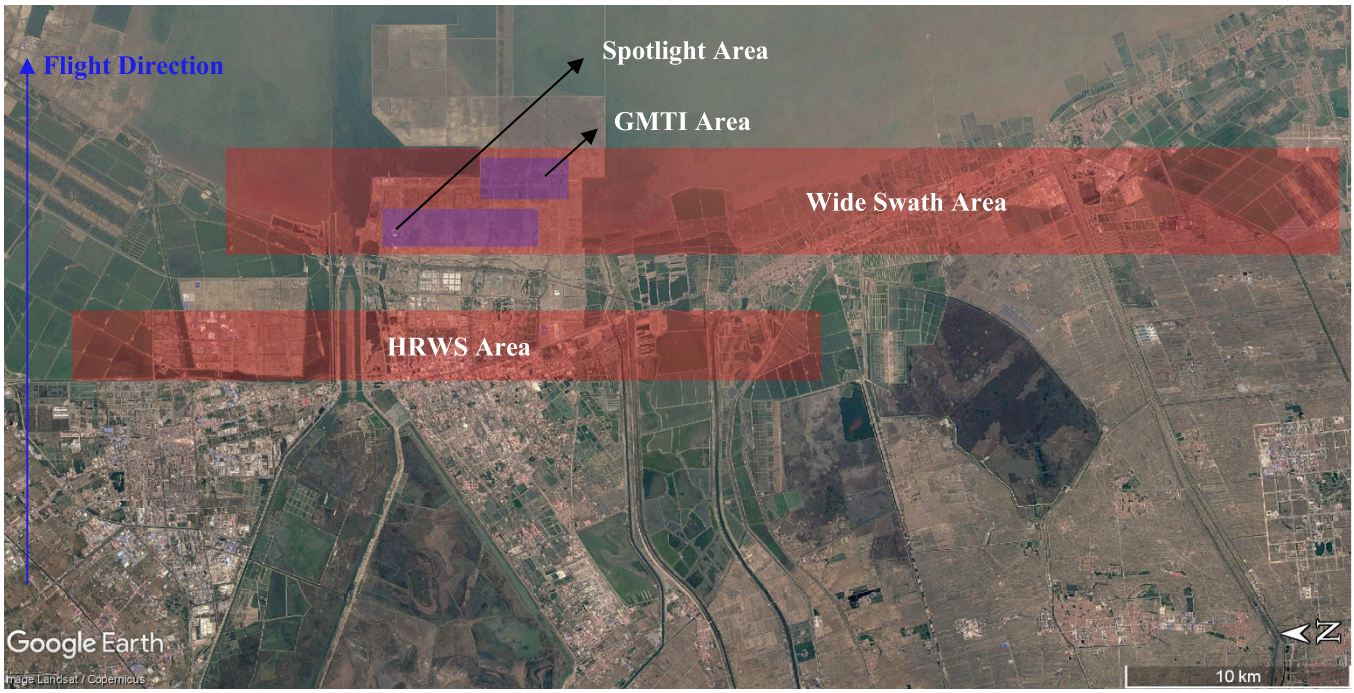


Fig. 9. Area covered by the airborne MIMO SAR experiments.

receiving beams of the four subapertures, both the wide swath and the spotlight echoes can be received by the nonsteering four subapertures. For the HRWS imaging mode, there is no steering.

By employing the STC scheme, the MIMO transceivers, and the phased array, it becomes feasible to realize the multimodal operation and the HRWS imaging. However, an important issue may arise with regard to the system PRF.

As aforementioned, the system PRF must be larger than the sum of the Doppler bandwidths of the two performed modes. Then, with the same level of ambiguities, one could also realize multiple SAR modes through toggling the pulses between PRIs. Following this concept, one single transmitter is enough. The STC scheme and the multiple transmitters seem unnecessary. To avoid this confusion, advantages of the proposed approach over the toggling scheme are analyzed in terms of the SNR.

Assume that the PRF of the proposed approach is the same as that of the toggling scheme operating at a doubled PRF. Then, the system PRIs of both schemes are identical. However, due to the interleaving of the multimodal pulses, the individual PRIs of the two modes performed by the toggling scheme are actually twice the system PRI (see Fig. 6). Based on this fact, the SNRs of the two schemes will be analyzed in the following.

For the proposed approach, the pulses of Mode1 and Mode2 are transmitted through Tx1 and Tx2, respectively. Then, the average transmitting powers of the two

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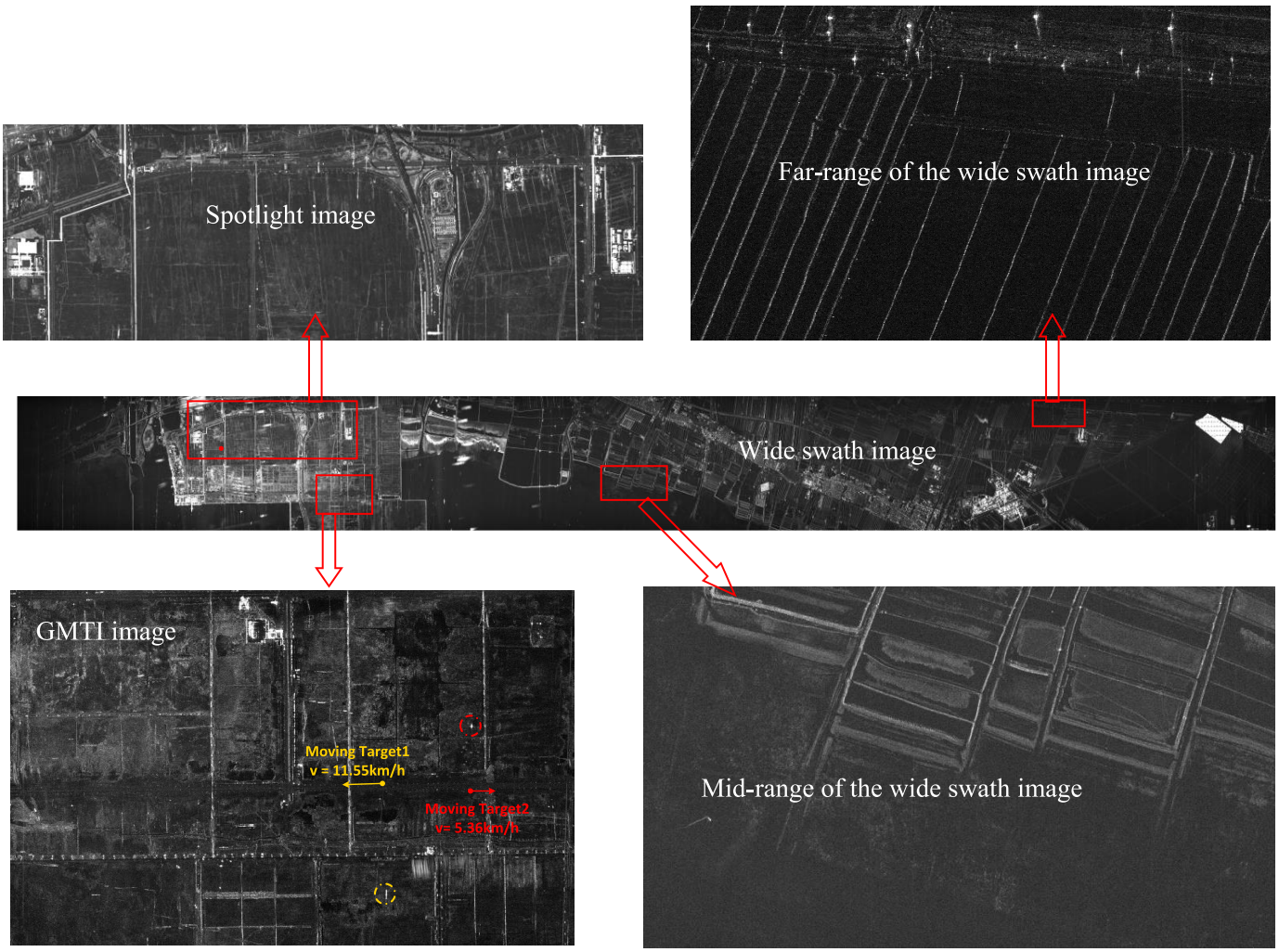


Fig. 10. MIMO SAR images of the simultaneously performed wide swath, spotlight, and GMTI operations. The corner reflector, which is used to compare the imaging performance of the wide swath and the spotlight modes, is denoted by a red point in the middle image. In this experiment, while the wide swath mode is used to observe a vast area to grasp the overall situation, the spotlight mode can be used to focus on the targets of interest at the same time. Furthermore, two moving targets are detected with a velocity of 11.55 and 5.36 km/h.

modes are

$$\bar{P}_{\mathrm{STC-Mode1}} = P \cdot \frac{T_1}{\mathrm{PRI}_{\mathrm{STC}}}$$
$$\bar{P}_{\mathrm{STC-Mode2}} = P \cdot \frac{T_2}{\mathrm{PRI}_{\mathrm{STC}}} \quad (5)$$

where T_1 and T_2 are the signal durations of Mode1 and Mode2, respectively. The peak power of both Tx1 and Tx2 is P.

For the toggling scheme, the pulses of Mode1 and Mode2 are transmitted by a single transmitter (Tx1 or Tx2). Then, the average transmitting power of the two modes is

$$\bar{P}_{\mathrm{toggle-Mode1}} = P \cdot \frac{T_1}{\mathrm{PRI}_{\mathrm{toggle-Mode1}}} = P \cdot \frac{T_1}{2 \cdot \mathrm{PRI}_{\mathrm{STC}}}$$
$$\bar{P}_{\mathrm{toggle-Mode2}} = P \cdot \frac{T_2}{\mathrm{PRI}_{\mathrm{toggle-Mode2}}} = P \cdot \frac{T_2}{2 \cdot \mathrm{PRI}_{\mathrm{STC}}}. \quad (6)$$

From (5) and (6), it can be concluded that the proposed approach offers better SNRs for both the modes.

However, if all the subarrays are reconfigured as one single transmitter to perform the toggling scheme, the average transmitting power of the two modes will be

$$\bar{P}_{\mathrm{toggle-Mode1}} = 2P \cdot \frac{T_1}{\mathrm{PRI}_{\mathrm{toggle-Mode1}}} = P \cdot \frac{T_1}{\mathrm{PRI}_{\mathrm{STC}}}$$
$$\bar{P}_{\mathrm{toggle-Mode2}} = 2P \cdot \frac{T_2}{\mathrm{PRI}_{\mathrm{toggle-Mode2}}} = P \cdot \frac{T_2}{\mathrm{PRI}_{\mathrm{STC}}}. \quad (7)$$

In this case, the average transmitting powers of the two modes are the same as those in (5). However, the Doppler bandwidth is halved due to the doubled transmitter length. Then, for HRWS imaging, the azimuthal resolution will be degraded. For multimodal operation, the halved Doppler bandwidth will reduce the number of looks in the multilook processing of the wide swath mode and, as a result, degrades the SNR.

Moreover, the STC-based airborne MIMO SAR intends to lay a foundation for transferring APC technologies to the spaceborne MIMO SAR sensors. The presented airborne MIMO SAR system, along with the theoretical analysis and the experimental results (detailed in Section IV), will enable a further investigation of the APC technologies to reduce the

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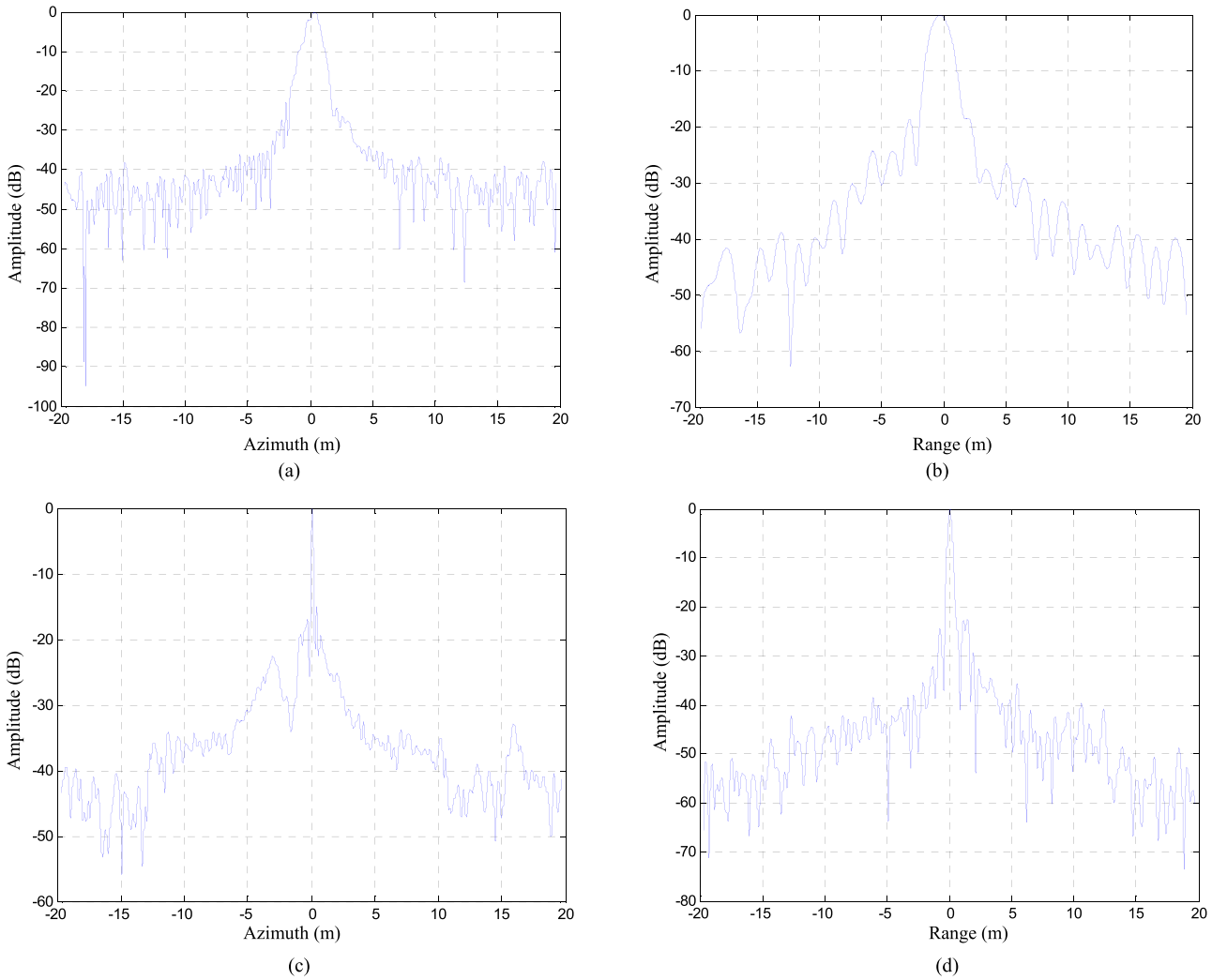


Fig. 11. Processing results of the corner reflector. (a) and (b) are the azimuth and range slices, respectively, of the corner reflector in the wide swath mode. (c) and (d) are the azimuth and range slices, respectively, of the corner reflector in the spotlight mode.

increased PRF and thus make it possible to realize spaceborne HRWS imaging and multimodal operation.

IV. Experimental Flight Results

The flight experiments of the multimodal operation are presented in this section. The phased array is installed at the aircraft bottom along the azimuthal direction (see Fig. 7). The parameters and performances are listed in Table I. In particular, the bandwidths used for the spotlight and the wide swath modes are located at the baseband and high band of the Doppler frequency domain, respectively. Since the echoes can be separated by bandpass filtering in the range-Doppler domain, the wide swath and spotlight modes can be performed simultaneously, with -35-dB intrainterferences (see Fig. 8). The area for the airborne MIMO SAR experiments is shown in Fig. 9. In the flight experiments, the HRWS mode and the multimodal operation mode were performed in two flight passes. We tried to cover the same area. However, the flight result was that the covered areas were adjacent. The experimental results of the multimodal operation are shown in Figs. 10 and 11.

According to the results, we can conclude that the resolution and swath width of the wide swath mode are 0.91 m (azimuth) $\times$ 1.54 m (range) and 50.35 km, respectively. The spotlight resolution is 0.15 m (azimuth) $\times$ 0.29 m (range). The performance results of the multimodal operation can be found in the corner-reflector-provided range and azimuth slices (see Fig. 11). Note that the theoretical azimuthal resolution of the wide swath mode is 0.23 m. However, to improve the SNR, this mode has been multilook-processed. For the GMTI mode, the radial velocities of two cooperative ground moving targets, which are computed as -3.13 and 1.69 m/s by differential global positioning system (GPS) processing, are measured as -3.21 and 1.49 m/s. The measurement errors are 0.08 and 0.2 m/s. Note that 1.69 m/s is the minimum detected velocity in the experiment. It should be distinguished from the theoretical minimum detectable velocity (MDV) [50], [51].

To offer a comparison, the HRWS mode is performed as well. Specifically, the chirps transmitted by Tx1 and Tx2 are located at the baseband and high band of the Doppler frequency domain, respectively. The configuration of the phased array is basically the same as that in the multimodal operation.

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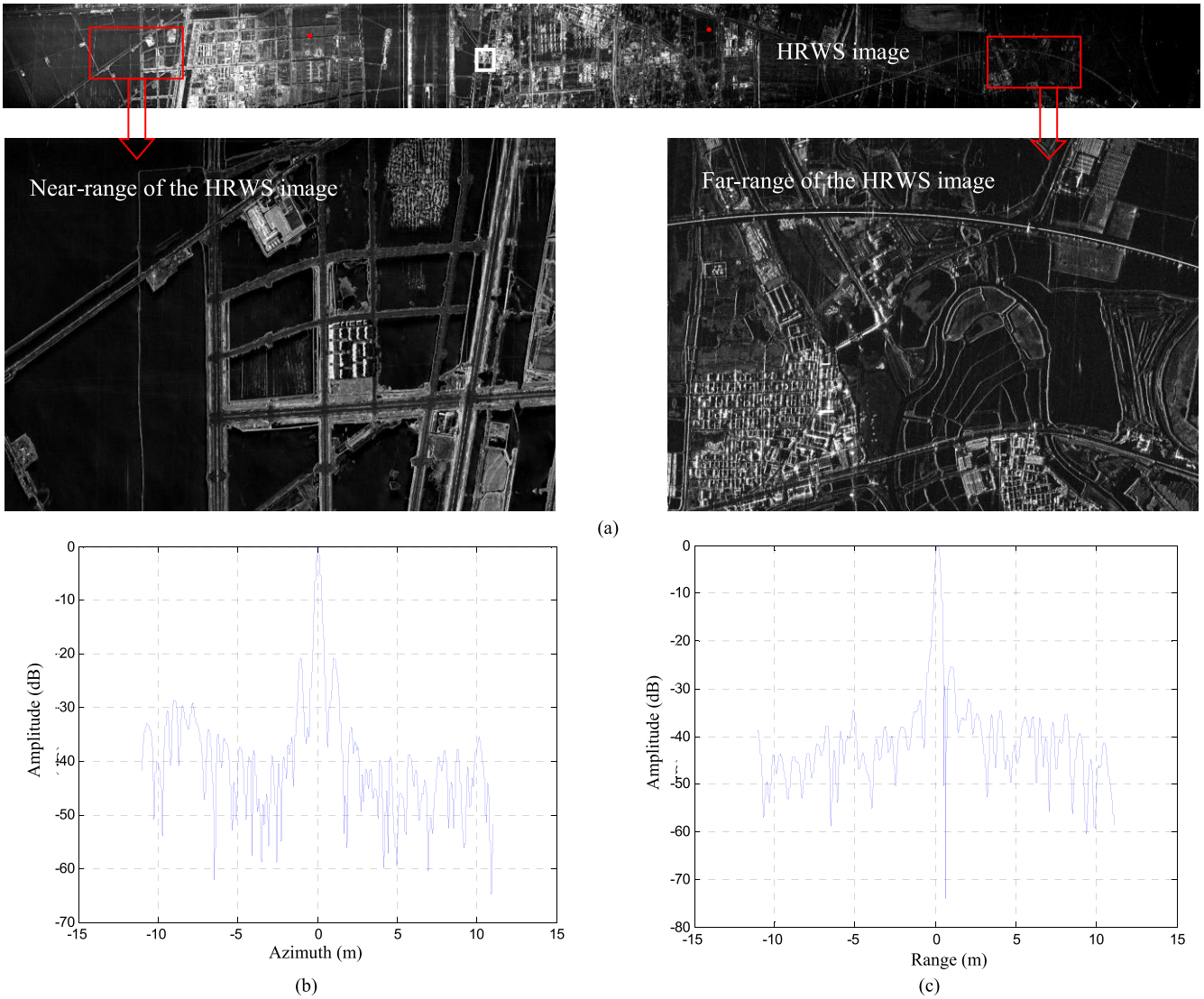


Fig. 12. HRWS imaging results. (a) MIMO SAR images. The red points denote the corner reflector. The white rectangle denotes the area of the SNR assessment of the distributed targets. This area is enlarged and analyzed in Fig. 13. (b) and (c) Azimuth and range slices of the red point, respectively.

TABLE II
PARAMETERS AND PERFORMANCES OF THE HRWS IMAGING MODE

Bandwidth	560 MHz
Duration	75 μs
Doppler Bandwidth	648 Hz
Range Resolution	0.29 m
Azimuthal Resolution	0.26 m
Swath Width	32.9 km

The main difference is that there is no steering for the HRWS imaging mode. The system parameters are the same as those listed in Table I. The individual parameters, including the bandwidth and duration, are listed in Table II. The performances are also listed in Table II. The experimental results are shown in Fig. 12.

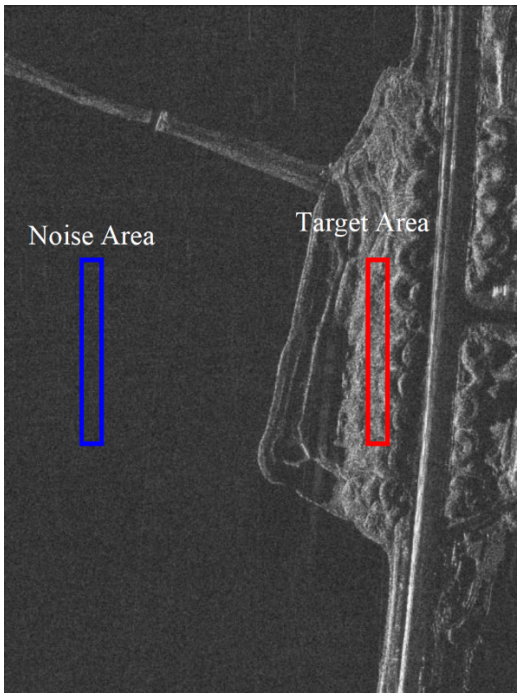
We can conclude from the experimental results that the resolution and swath width are 0.26 m (azimuth) × 0.29 m (range) and 32.9 km, respectively. The achieved HRWS imaging performances can be found in the corner-reflector-provided range and azimuth slices [see Fig. 12(b) and (c)].

Theoretically speaking, with two transmitters and four receivers, we can obtain eight equivalent transceiver channels for the HRWS mode. Since each equivalent transceiver channel would produce one image, we can obtain eight images. Henceforth, if compared with one of the eight images, the SNR can be improved by 9 dB through synthesis of the eight images. This SNR comparison is one between the conventional single-channel SAR and the MIMO SAR systems. In particular, if a conventional single-channel SAR aims to realize the same azimuthal resolution and swath width, its antenna should be as large as the MIMO subantenna. In this case, the transmitting power and gain of the single-channel SAR become equal to those of the MIMO subantennas. Consequently, for the image produced by the conventional single-channel SAR, its SNR is equal to that in each of the eight MIMO images. Then, by synthesizing the eight MIMO images, the SNR can be improved by 9 dB. However, due to the impact of systematic errors, the SNR is actually increased by 7 dB for the point target and 6 dB for the distributed targets (see Fig. 13). Consequently, with the enhanced SNR,

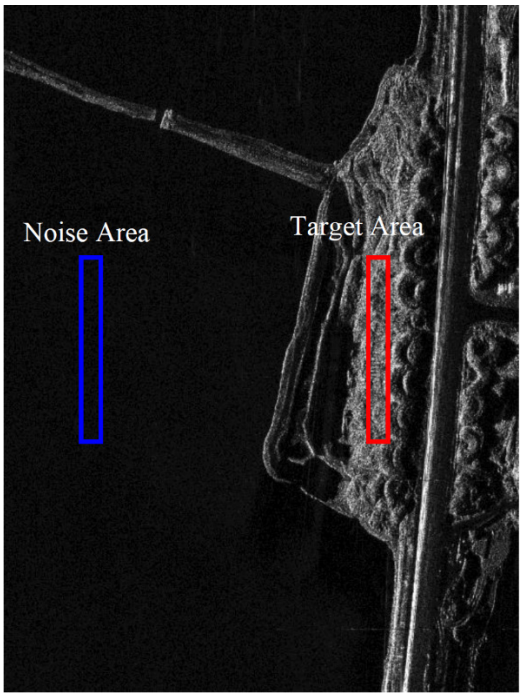
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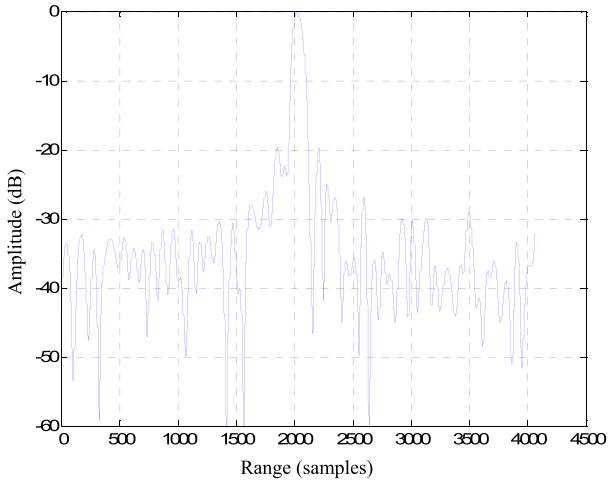
(a)



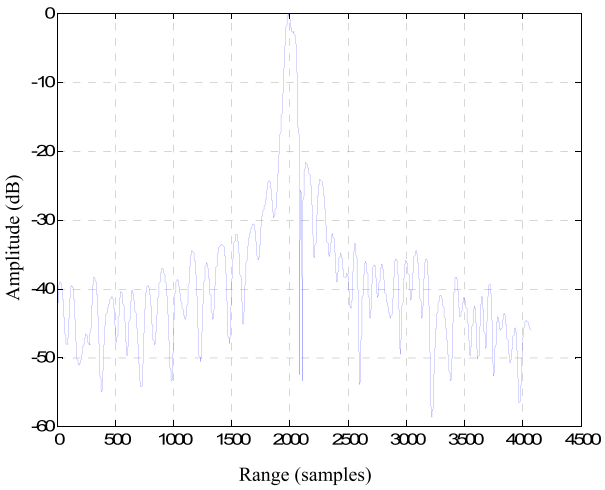
(b)



(c)



(d)



(e)

Fig. 13. SNR comparison between the single-channel image and the MIMO SAR image. (a) Single-channel image. The white rectangle denotes the area for the SNR assessment of the distributed targets. The red point, which denotes the corner reflector, is used for the SNR assessment of the point target. (b) and (c) Enlarged views of the white rectangular areas in (a) and Fig. 12(a). The blue rectangles in (b) and (c) denote the noise areas for the SNR computation. The red rectangular areas in (b) and (c) denote the distributed targets for the SNR computation. It can be computed that the SNR in (b) is 9.41 dB and that the SNR in (c) is 15.60 dB. The SNR improvement for the distributed targets is, as a result, 6.19 dB. (d) and (e) Range slices of the same corner reflectors in (a) and Fig. 12(a), respectively. According to (d) and (e), the SNR improvement for the point target is approximately 7 dB.

it becomes feasible for the MIMO SAR to realize airborne HRWS imaging.

It can be concluded from the experimental results of the HRWS mode that the proposed MIMO SAR system is capable

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TABLE III
PARAMETERS OF THE F-SAR AT C-BAND [54]

Carrier Frequency	5.3 GHz
Bandwidth	400 MHz
System PRF	5000Hz
Transmitting Power	2200W
Range Resolution	0.6 m
Azimuthal Resolution	0.3 m
Swath Width	12.5 km

of precisely observing every target in a vast area. However, as mentioned above, there is no need to image all the targets with high resolution for certain applications. In this case, the multimodal operation of wide swath imaging, spotlight, and GMTI is crucial. According to the experimental results of the multimodal operation, we can conclude that our proposed system can cover a vaster area and provide a higher resolution for the targets of interest. The highest achievable swath width and resolution are superior to those of the HRWS mode. In addition, the MIMO SAR system is able to realize a more sensitive GMTI mode by exploiting the MIMO diversities [52], [53]. As a consequence, the efficiency of the multimodal operation is optimal. Note that although the HRWS mode and the multimodal operation mode covered different areas in Fig. 9, the experimental images, detailed resolutions, and swath widths in Tables I and II are sufficient for the comparison.

The essential advantages of the MIMO SAR concepts are achieved by dividing the phased array into multiple subapertures. With smaller subapertures, we can achieve wider coverage and higher azimuth resolution. Moreover, the multiple equivalent transceiver channels are capable of improving the SNR and reducing the system PRF. For the airborne geometry, the imaging performances are mainly limited to the SNR. From the viewpoint of SNR, the introduced MIMO SAR is qualified for the airborne HRWS imaging and multimodal operation. If compared with F-SAR (see Table III), the performances of the introduced MIMO SAR system are far more advanced.

However, the technologies presented in this article cannot be directly transferred to the spaceborne MIMO SAR system. If one tries to realize HRWS imaging and multimodal operation on the spaceborne MIMO SAR system, the APC schemes should be further investigated to reduce the PRF.

V. CONCLUSION AND FUTURE DIRECTIONS

Due to the potential capability of realizing HRWS imaging, research on the MIMO SAR system has increased over the past few decades. In this article, it is further proposed to realize the multimodal operation of wide swath imaging, spotlight, and GMTI. With the abilities of HRWS imaging and multimodal operation, the airborne MIMO SAR system is qualified for various future SAR missions. The developed airborne MIMO SAR system can perform the HRWS imaging mode to realize a wider swath and a higher geometric resolution simultaneously. In addition, if only a sparse set of targets in a vaster area are of interest, the MIMO radar can perform multimodal operation. In particular, whereas the wide swath mode is used to grasp the overall situation, the spotlight and GMTI modes can be simultaneously formed to precisely observe the targets of interest. The key behind the feasibility of the airborne MIMO experiments is the STC waveform scheme. With the STC waveform scheme, the intramodal interferences introduced by the simultaneous transmissions can be suppressed down to -35 dB. Thus, the HRWS imaging mode and the multimodal operation concept can be performed in the airborne MIMO SAR system. The content of this article has been validated by flight experiments. In the future, the STC scheme will be improved to reduce the doubled PRF and henceforth to realize spaceborne HRWS imaging and multimodal operation.

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REFERENCES

[1] I. G. Cumming and F. H. Wong, *Digital Processing of Synthetic Aperture Radar Data: Algorithms and Implementation*. Boston, MA, USA: Artech House, 2005.
[2] J. C. Curlander and R. N. McDonough, *Synthetic Aperture Radar Systems and Signal Processing*. New York, NY, USA: Wiley, 1991.
[3] A. Currie and M. A. Brown, "Wide-swath SAR," *IEE Proc. F Radar Signal Process.*, vol. 139, no. 2, pp. 122–135, Apr. 1992.
[4] F. Ulaby and D. Long, *Microwave Radar and Radiometric Remote Sensing*. Ann Arbor, MI, USA: Univ. Michigan Press, 2014.
[5] S. Huber, F. Q. de Almeida, M. Villano, M. Younis, G. Krieger, and A. Moreira, "Tandem-L: A technical perspective on future spaceborne SAR sensors for Earth observation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4792–4807, Aug. 2018.
[6] M. Villano, G. Krieger, M. Jager, and A. Moreira, "Staggered SAR: Performance analysis and experiments with real data," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6617–6638, Nov. 2017.
[7] G. Krieger *et al.*, "MirrorSAR: A fractionated space radar for bistatic, multistatic and high-resolution wide-swath SAR imaging," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 149–152.
[8] J.-H. Kim, M. Younis, A. Moreira, and W. Wiesbeck, "Spaceborne MIMO synthetic aperture radar for multimodal operation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2453–2466, May 2015.
[9] N. Gebert, G. Krieger, and A. Moreira, "Digital beamforming on receive: Techniques and optimization strategies for high-resolution wide-swath SAR imaging," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 45, no. 2, pp. 564–592, Apr. 2009.
[10] G. Krieger and A. Moreira, "Potential of digital beamforming in bi- and multistatic SAR," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Jul. 2003, pp. 527–529.
[11] M. Younis, S. Huber, A. Patyuchenko, F. Bordoni, and G. Krieger, "Performance comparison of reflector- and planar-antenna based digital beam-forming SAR," *Int. J. Antennas Propag.*, vol. 2009, pp. 1–13, Jun. 2009.
[12] A. Freeman *et al.*, "SweepSAR: Beam-forming on receive using a reflector-phased array feed combination for spaceborne SAR," in *Proc. IEEE Radar Conf.*, May 2009, pp. 1–9.
[13] G. Krieger *et al.*, "Advanced digital beamforming concepts for future SAR systems," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2010, pp. 245–248.
[14] M. Younis, C. Fischer, and W. Wiesbeck, "Digital beamforming in SAR systems," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 7, pp. 1735–1739, Jul. 2003.
[15] K. Tomiyasu, "Tutorial review of synthetic-aperture radar (SAR) with applications to imaging of the ocean surface," *Proc. IEEE*, vol. 66, no. 5, pp. 563–583, May 1978.
[16] W.-Q. Wang, "Space–time coding MIMO-OFDM SAR for high-resolution imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 8, pp. 3094–3104, Aug. 2011.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:33:14 UTC from IEEE Xplore. Restrictions apply.

[17] Q. Wu, M. Xing, Z. Bao, and H. Shi, "Wide swath, high range resolution imaging with MIMO-SAR," in *Proc. IET Conf. Publications*, Apr. 2009, pp. 1–6.
[18] G. Krieger, "MIMO-SAR: Opportunities and pitfalls," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2628–2645, May 2014.
[19] M. Younis, G. Krieger, and A. Moreira, "MIMO SAR techniques and trades," in *Proc. EuRAD*, Nuremberg, Germany, Oct. 2013, pp. 141–144.
[20] W.-Q. Wang, "Large-area remote sensing in high-altitude high-speed platform using MIMO SAR," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 5, pp. 2146–2158, Oct. 2013.
[21] J. H. G. Ender and J. Klare, "System architectures and algorithms for radar imaging by MIMO-SAR," in *Proc. IEEE Radar Conf.*, May 2009, pp. 1–6.
[22] G. Krieger *et al.*, "MIMO-SAR and the orthogonality confusion," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 1533–1536.
[23] E. Fishler, A. Haimovich, R. Blum, D. Chizhik, L. Cimini, and R. Valenzuela, "MIMO radar: An idea whose time has come," in *Proc. IEEE Radar Conf.*, Apr. 2004, pp. 71–78.
[24] W.-Q. Wang, "MIMO SAR imaging: Potential and challenges," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 28, no. 8, pp. 18–23, Aug. 2013.
[25] J. Wang, X. Liang, C. Ding, L. Chen, Z. Wang, and K. Li, "A novel scheme for ambiguous energy suppression in MIMO-SAR systems," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 2, pp. 344–348, Feb. 2015.
[26] J. X. D. Wang Liang *et al.*, "Multi-input multi-output frequency-modulated continuous wave synthetic aperture radar system using beat-frequency division waveforms," *Meas. Sci. Technol.*, vol. 24, no. 7, pp. 74016–74024, Jul. 2013.
[27] J. Wang, X.-D. Liang, L.-Y. Chen, L.-N. Wang, and K. Li, "First demonstration of joint wireless communication and high-resolution SAR imaging using airborne MIMO radar system," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6619–6632, Sep. 2019.
[28] A. Paulraj and T. Kailath, "Increasing capacity in wireless broadcast systems using distributed transmission/directional reception," U.S. Patent. 5 345 599, Sep. 6, 1994.
[29] E. Fishler, A. Haimovich, R. S. Blum, L. J. Cimini, D. Chizhik, and R. A. Valenzuela, "Spatial diversity in radars—Models and detection performance," *IEEE Trans. Signal Process.*, vol. 54, no. 3, pp. 823–838, Mar. 2006.
[30] K. W. Forsythe and D. W. Bliss, *MIMO Radar: Concepts, Performance Enhancements, and Applications. MIMO Radar Signal Processing*. Hoboken, NJ, USA: Wiley, 2015.
[31] D. R. Fuhrmann and G. San Antonio, "Transmit beamforming for MIMO radar systems using signal cross-correlation," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 44, no. 1, pp. 171–186, Jan. 2008.
[32] D. Cristallini, D. Pastina, and P. Lombardo, "Exploiting MIMO SAR potentialities with efficient cross-track constellation configurations for improved range resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 1, pp. 38–53, Jan. 2011.
[33] J. Ender, "Along-track array processing for MIMO-SAR/MTI," in *Proc. Eur. Conf. Synth. Aperture Radar*, Friedrichshafen, Germany, vol. 4, Jun. 2008, pp. 67–70.
[34] B. Zou, Z. Dong, and D. Liang, "Design and performance analysis of orthogonal coding signal in MIMO-SAR," *Sci. China Inf. Sci.*, vol. 54, no. 8, pp. 1723–1737, Aug. 2011.
[35] D. Cerutti-Maori, I. Sikaneta, J. Klare, and C. H. Gierull, "MIMO SAR processing for multichannel high-resolution wide-swath radars," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 5034–5055, Aug. 2014.
[36] J. Ender, "MIMO-SAR," in *Proc. Int. Radar Symp.*, Cologne, Germany, 2007, pp. 667–674.
[37] P. Woodward, "Radar ambiguity analysis," RRE, New York, NY, USA, Tech. Note 731, Feb. 1967.
[38] G. Krieger, N. Gebert, and A. Moreira, "Multidimensional waveform encoding: A new digital beamforming technique for synthetic aperture radar remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 1, pp. 31–46, Jan. 2008.
[39] T. Rommel *et al.*, "An orthogonal waveform scheme for imaging MIMO-Radar applications," in *Proc. IRS*, Dresden, Germany, Jun. 2013, pp. 917–922.
[40] J.-H. Kim, M. Younis, A. Moreira, and W. Wiesbeck, "A novel OFDM chirp waveform scheme for use of multiple transmitters in SAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 3, pp. 568–572, May 2013.
[41] J. Wang, X. Liang, C. Ding, L. Chen, L. Zhou, and Y. Dong, "An improved OFDM chirp waveform used for MIMO SAR system," *Sci. China Inf. Sci.*, vol. 57, no. 6, pp. 1–9, Jun. 2014.
[42] J. Wang, L.-Y. Chen, X.-D. Liang, C.-B. Ding, and K. Li, "Implementation of the OFDM chirp waveform on MIMO SAR systems," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 5218–5228, Sep. 2015.
[43] J.-H. Kim, A. Ossowska, and W. Wiesbeck, "Investigation of MIMO SAR for interferometry," in *Proc. Eur. Radar Conf.*, Oct. 2007, pp. 51–54.
[44] F. He, Z. Dong, and D. Liang, "A novel space-time coding Alamouti waveform scheme for MIMO-SAR implementation," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 2, pp. 229–233, Feb. 2015.
[45] J. Wang, X.-D. Liang, L.-Y. Chen, and K. Li, "A novel space–time coding scheme used for MIMO-SAR systems," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1556–1560, Jul. 2015.
[46] C. Z. Meng *et al.*, "MIMO-SAR waveform separation based on inter-pulse phase modulation and range-Doppler decouple filtering," *Electron. Lett.*, vol. 49, no. 6, pp. 420–422, Mar. 2013.
[47] J. Dall and A. Kusk, "Azimuth phase coding for range ambiguity suppression in SAR," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Sep. 2004, pp. 1734–1737.
[48] F. Bordoni, M. Younis, and G. Krieger, "Ambiguity suppression by azimuth phase coding in multichannel SAR systems," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 617–629, Feb. 2012.
[49] F. Zhou, J. Ai, Z. Dong, J. Zhang, and M. Xing, "A novel MIMO-SAR solution based on azimuth phase coding waveforms and digital beamforming," *Sensors*, vol. 18, no. 10, pp. 1–16, Oct. 2018.
[50] W. Koch, "Effect of Doppler ambiguities on GMTI tracking," in *Proc. IEE RADAR*, Edinburgh, U.K., 2002, pp. 153–157.
[51] J. Qian, "Study on methods for SAR-GMTI," Ph.D. dissertation, Xidian Univ., Xi'an, China, 2011.
[52] X. J. Zhang, "Study on SAR/GMTI Technology for multi-phase centre receive system," Ph.D. dissertation, Nanjing Univ. Aeronaut. Astronaut., Nanjing, China, 2009.
[53] J. H. G. Ender, P. Berens, A. R. Brenner, L. Rossing, and U. Skupin, "Multi-channel SAR/MTI system development at FGAN: From AER to PAMIR," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jun. 2002, pp. 1697–1701.
[54] R. Horn, A. Nottensteiner, A. Reigber, J. Fischer, and R. Scheiber, "F-SAR—DLR's new multifrequency polarimetric airborne SAR," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2009, pp. II-902–II-905.



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